3.(i) 1st quadrant (ii) 3rd quadrant (iii) 2nd quadrant.

4.(i) $\frac{-\sqrt{3}}{2}$ (ii) $\frac{-\sqrt{3}}{2}$ (iii) $\frac{2}{\sqrt{3}}$ (iv) 1

(v) $\sqrt{2}$ 10. $\frac{-7}{2}$

Exercise 4.2

1.(i) $\frac{2\sqrt{2}}{\sqrt{3}-1}$ (ii) $-\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)$ (iii) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$

2.(i) $\sin 92^\circ$ (ii) $\frac{\sqrt{3}}{2}$ (iii) $\cos 80^\circ$ (iv) $\frac{\sqrt{3}}{2}$

3.(i) $\frac{-33}{65}$ (ii) $\frac{-16}{65}$ (iii) $\frac{16}{33}$

6. $\frac{2}{11}$, $\alpha+\beta$ lies in 1st quadrant. 7. $\pm\sqrt{2}-1$

9.(i) $\frac{9}{13}$ (ii) $-\frac{828}{2197}$ 10. $-\frac{44}{125}, -\frac{117}{44}$ 14. $\sqrt{2}-1$

Exercise 4.3

1.(i) $\frac{1}{2}\left(\cos\frac{A}{4}-\cos\frac{A}{2}\right)$ (ii) $\frac{1}{2}\left(-\sin 2A+\frac{\sqrt{3}}{2}\right)$

(iii) $\frac{1}{2}\left(\sin 4A-\sin\frac{2A}{3}\right)$ (iv) $\frac{1}{2}(\sin 10\theta-\sin 4\theta)$

2.(i) $2\sin\frac{3A}{2}\cos\frac{A}{2}$ (ii) $2\cos 3A\cos A$

(iii) $2\sin 2\theta\cos 4\theta$ (iv) $-2\sin\frac{3\theta}{2}\sin\frac{\theta}{2}$

Exercise 4.4

1.(i) $\frac{-\pi}{6}$ (ii) $\frac{-\pi}{4}$ (iii) $\frac{\pi}{6}$ (iv) $\frac{3\pi}{4}$

4. $x=\frac{1}{6}$ 5. $x=\frac{1}{2}$

6.(i) $\frac{4}{5}$ (ii) $\frac{1}{\sqrt{10}}$ 7. $\frac{-33}{65}$ 10. $\frac{\pi}{4}+\frac{x}{2}$

Exercise 4.5

1	2	3	4	5	6	7	8	9	10	11	12	13
(b)	(a)	(c)	(b)	(b)	(d)	(a)	(d)	(a)	(c)	(c)	(d)	(b)
14	15	16	17	18	19	20	21	22	23	24	25	
(c)	(b)	(a)	(c)	(b)	(c)	(c)	(b)	(c)	(a)	(b)	(d)	

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